Nishal Stanislaus Silva, *PhD*

Machine Learning Engineer | Applied AI, MLOps, Evaluation Design | Cross-Functional Delivery

Canadian Permanent Resident, Eligible to work in Canada without sponsorship

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19th October 2026

Re: AI / ML Product Engineer – TD

Dear Hiring Manager,

I am applying for the AI / ML Product Engineer role in Analytics, Insights & Artificial Intelligence at TD. I am a Machine Learning Engineer with a PhD (University of Trento, 2025) and 10+ years across industry and academia. I build ML systems end-to-end, and much of my industry work was turning vague business requests into specified, measurable solutions with operations and engineering stakeholders. I understand this is a leadership and enablement role more than a hands-on build role, and the part that fits me is defining the task, setting how success is measured, and helping engineers ship it.

Three parts of my work match your posting. For evaluation, I design benchmark suites with baselines and ablations, and I released four open datasets on Zenodo. My published real-time system runs at 14 ms on a Raspberry Pi 4, 74x faster than the DTW baseline (Journal of the Audio Engineering Society, 2026), and I owned its full lifecycle from data acquisition to deployment and monitoring. For stakeholders in a regulated setting, at Forestpin I worked with compliance and risk teams to turn detection requirements into technical specs for financial forensics pipelines. For enablement, at MAS Holdings I mentored engineers, ran group-wide computer vision training seminars and introduced code review and CI/CD, and I supervised two BSc theses at Trento.

I should be direct about the gaps. My generative AI work is applied: I use LLMs, RAG and Claude Code daily, but I have not shipped an LLM or agent product at scale, and I have not used LangChain, LlamaIndex or Semantic Kernel in production. I have not held a formal product owner role or managed a Jira backlog, and I have no banking experience. My cloud work is on AWS (S3, ECS, MWAA, RDS) with Airflow and Docker, not Azure. I expect to pick these up quickly, and I would aim to start with clear acceptance criteria and evaluation sets for the first AI initiatives.

I am a Canadian permanent resident based in Toronto, ON, eligible to work without sponsorship, and available immediately for the Toronto office. I would welcome the opportunity to discuss how my experience in applied ML, evaluation and mentoring can support the Analytics, Insights & AI team at TD. My portfolio, publications and open datasets are available at nishal.xyz.

Sincerely,

Nishal Stanislaus Silva, PhD